

Chengzhu Duan

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EDUCATION

University of California, San Diego
Computer Science, B.S.
GPA: 3.80 / 4.00 (Cum Laude)

La Jolla, CA
Sept. 2017 – Mar. 2021

WORK EXPERIENCE

Coinbase

Senior Software Engineer

New York, NY
Apr. 2021 – Present

- Pioneered a **reusable adapter framework** for integrating venue APIs, expanding liquidity access for institutional clients on [Coinbase Prime](#)
- Built scalable and fault-tolerant Brokerage services in **Go**, supporting **billions in trading volume** across retail and institutional order flows
- Drove the org-wide roadmap for high-fidelity market data, delivering updates for **250k+** trading pairs across asset classes within **10ms** latency
- Led the zero-to-one launch of Coinbase Equities trading, enabling support for **stocks, ETFs, and options**

Coinbase

Software Engineer Intern

San Francisco, CA
Jun. 2020 – Sept. 2020

- Built billing, invoice, and email notification services in **Ruby on Rails** for [Coinbase Prime](#)
- Automated invoice generation and **AWS S3** storage, saving finance and operations teams **40 hours/week** and recovering **\$350K** in unpaid invoices

Omniscience Corporation

Software Engineer Intern

Palo Alto, CA
Jul. 2019 – Sept. 2019

- Built underwriting automation services with **Flask**, **Swagger Codegen**, and **SQLAlchemy** for insurance workflows used by enterprise clients
 - Implemented document upload, sharing, editing, and labeling workflows with **Angular**, **Spring Boot**, and **AWS S3**
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PROJECTS

Swag-Bot

</> Node.js, Mongoose, MySQL

★ 115 ♡ 70
Jul. 2018 – Jun. 2019

- Built an open-source Discord automation platform used across **15** servers with **500+** users
- Designed persistence and analytics workflows with **Node.js**, **MySQL**, and **MongoDB** for point systems, moderation, and media commands

Building Sensor Data Search Engine

</> Angular, Flask, Node.js, MySQL, Pytorch, Docker, C++, RPC

📄 [ACM BuildSys'21 Paper](#)
Jan. 2020 – Nov. 2021

- Built time-series search and visualization workflows for terabytes of IoT sensor data using **Angular**, **Flask**, **Node.js**, and **MySQL**
- Engineered query evaluation with **Sliding Window Feature Extraction**, **Locality-Sensitive Hashing**, and heuristic **Dynamic Time Warping**
- Searched across **20 million** data points in under **3 seconds**, with results published in [ACM BuildSys'21](#)

Your Clear Way Mobile App

</> AWS AppSync/Amplify/Lambda/DynamoDB, Node.js, GraphQL, React Native, Expo

🍃 Non-profit Work
Jan. 2021 – Mar. 2022

- Led a team of **7** developers building a React Native app for homeowners to track eco-friendly home upgrades and earn progress badges
 - Architected point and badge services on **AWS Amplify**, using GraphQL, authentication, serverless functions, and persistent storage
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SKILLS

Languages:

Go, Python, Ruby, TypeScript, Java, C/C++, SQL, LaTeX

Backend & Distributed Systems:

Kafka, gRPC, REST, Ruby on Rails, Node.js, Django, Flask

Cloud & Data:

AWS, PostgreSQL, MySQL, MongoDB, DynamoDB, Firebase

Domains:

Trading Systems, Market Data, Brokerage, Institutional Crypto, Equities

Leadership:

Technical Roadmaps, Cross-functional Execution, Mentorship